

Class 38 Fundamental Theorem of Calculus

Goals

1. Prove the Mean Value Theorem
2. Prove the Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus (FTC) makes the connection between derivatives and integrals. The first proofs were constructed independently by Isaac Newton and Gottfried Leibniz in the late 1600's. The proof we have today requires us to utilize several preliminary theorems.

1. Extreme Value Theorem:

2. Rolle's Theorem: If f is continuous on $[a,b]$ and differentiable on (a,b) and if $f(a) = f(b)$, then

Proof involves 3 cases and the Extreme Value Theorem

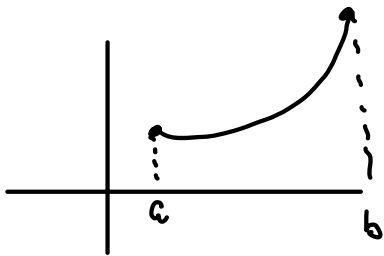
Case 1.

Case 2.

Case 3.

3. Mean Value Theorem: If f is continuous on $[a,b]$ and differentiable on (a,b) , then

Proof: Apply Rolle's Theorem to the function $h(x)$ that measures the gap between the graph of f and the line $g(x)$ joining the points $(a,f(a))$ and $(b,f(b))$.

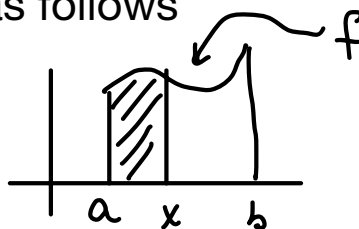


Verify that $h(x)$ satisfies all three conditions for Rolle's theorem so there must be at least one number c in (a,b) with $h'(c) = 0$.

4. Final fact before FTC: If two functions, f and g , have the same derivative then $f(x) = g(x) + C$ for some constant C .

Fundamental Theorem of Calculus

Part 1: If f is continuous on $[a,b]$ and x is any number in (a,b) , and if we define an accumulation function as follows



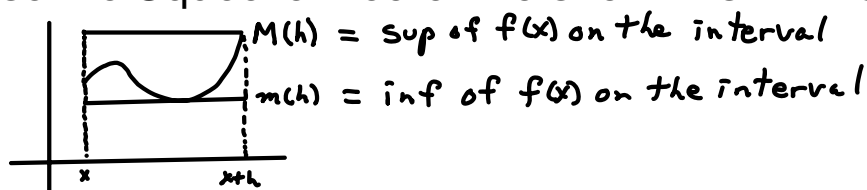
Then

Part 2: If f is continuous on $[a,b]$ and F is any anti derivative of f , then

Proofs:

For part 1 use the definition of derivative on the accumulation function $G(x)$.

Use the Squeeze theorem to show this limit is $f(x)$.

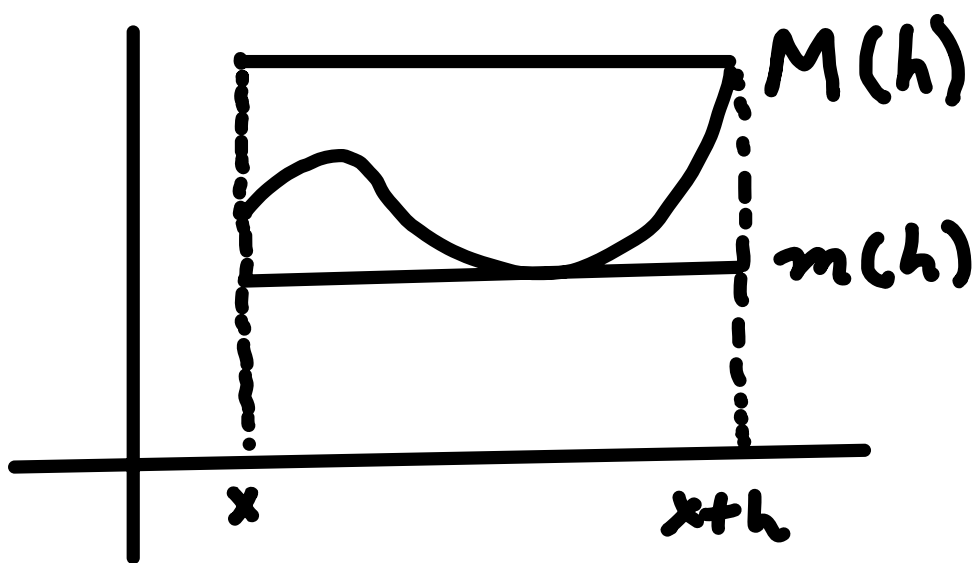


Let P_1 be the partition of $[x, x+h]$ into just one piece with width h .

But $L(f, P_1) =$

and $U(f, P_1) =$

So we get



Part 2 follows from Part 1 and the fact that if two functions have the same derivative they must differ by an added constant.

Let F be any antiderivative of f so $F' = f = G'$